Aarogya Blood Donation System: ML-Based Blood Requirement Prescription Verification

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1. Introduction

The Aarogya Blood Donation System is a NITJ campus platform designed to connect blood donors with patients in need. A key challenge in such systems is ensuring that only genuine blood requests are processed. Currently, users can upload any document (e.g., Aadhaar card, bills), which may lead to misuse.

To address this issue, this project proposes a Machine Learning (ML)-based document verification system that validates whether an uploaded PDF is a legitimate doctor-issued prescription indicating blood requirement.

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2. Problem Statement

Patients are required to upload a document as proof of blood requirement. However:

- Users may upload incorrect or unrelated documents
- Manual verification is time-consuming and unreliable
- There is a risk of misuse of the system

Therefore, an automated solution is required to verify the authenticity of uploaded documents before allowing donor matching.

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3. Objective

The objective of this project is to design and implement an automated system that:

- Extracts text from uploaded PDFs using OCR
- Classifies the document as:
  - Valid Prescription
  - Invalid Document

- Allows only verified requests to proceed to donor matching

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4. Found Research Papers

1. Paper Title: AI for Medical Informatics: The Application of Optical Character Recognition Technology in the Transcription of Pharmaceutical Labels (2025)

• What they did:
The study developed a mobile-based system that uses OCR to automatically read and extract information from pharmaceutical labels. It focuses on helping users (especially elderly patients) understand medication details such as drug name, dosage, frequency, and medical indications.

• Technique used:
The system follows a multi-stage pipeline including image acquisition, preprocessing (noise removal, contrast enhancement), OCR-based text extraction, and intelligent post-processing. Advanced techniques such as fuzzy matching using Levenshtein distance, regular expressions for dosage detection, dictionary-based validation, and Named Entity Recognition (NER) were used to improve accuracy and interpret medical information.

• Limitation:
The system's performance depends heavily on image quality, and OCR errors may occur due to blur, poor lighting, or complex label formats. It also faces challenges with multilingual support, handwriting recognition, and handling unstructured or degraded medical documents.

2. Paper Title: Improving Classification Accuracy for Unstructured Medical Documents via Multi-Engine OCR and Deep Learning Collaboration (2026)

• What they did:
The study proposed a system to automatically classify different types of medical documents (such as prescriptions, clinical notes, reports, and insurance forms) by combining multiple OCR engines and deep learning models. Instead of relying on a single OCR, the system dynamically selects the most suitable OCR engine based on document type and improves classification accuracy for real-world heterogeneous medical data.

• Technique used:
The system uses a multi-stage pipeline including document profiling, adaptive OCR engine selection (print-based, handwriting-based, and table-based OCR), and multi-engine text fusion. It

further applies ensemble deep learning models such as CNN, LSTM, and Transformer (BERT) for classification. Confidence-weighted voting and feature fusion techniques are used to combine outputs from multiple OCR engines and classifiers, improving overall accuracy.

• Limitation:
The system is computationally complex and requires higher processing resources due to multiple OCR engines and deep learning models. It is tested on a limited number of document categories and datasets, which may affect generalization. Additionally, the framework depends on accurate OCR selection and may struggle with extremely poor-quality or highly ambiguous documents.

3. Paper Title: Multipage Medical Document Classification of Printed and Scanned Pages Using Machine Learning Algorithms (2025)

• What they did:
The study developed an automated system to classify medical documents (such as prescriptions, lab reports, radiology reports, and consultation notes) from PDF files. The system converts documents into text, processes them, and categorizes them into predefined classes to reduce manual effort in healthcare and insurance workflows.

• Technique used:
The system uses OCR to convert PDF documents into text, followed by text preprocessing (cleaning and removing irrelevant words). Feature extraction is performed using Count Vectorizer and TF-IDF to convert text into numerical form. Machine learning algorithms such as Logistic Regression and Decision Tree are then applied for classification. Additionally, Chi-square feature selection is used to build a vocabulary of important words for each document category.

• Limitation:
The system relies heavily on the quality of OCR output, and errors in text extraction can reduce accuracy. It uses traditional machine learning methods, which may not capture complex document structures or context as effectively as deep learning models. It is also limited to predefined categories and may not generalize well to new or unseen document types.

4. Paper Title: Medical Document Classification

• What they did:
The study developed a system to automatically classify medical text documents into predefined categories such as cardiology, neurology, respiratory, and digestive. The system analyzes the

content of documents and assigns them to the most relevant category to reduce manual effort in hospitals.

• Technique used:
The system uses text mining techniques including tokenization, stop-word removal, and keyword extraction. It calculates term frequency (TF) of words to assign weights to different categories based on predefined keyword lists stored in a database. The classification is done by selecting the category with the highest weight score.

• Limitation:
The system works only on text data and does not handle images or PDFs directly (no OCR). It relies on predefined keyword lists, making it less flexible and not suitable for complex or unseen documents. It also uses basic techniques and lacks advanced models like TF-IDF or deep learning, limiting its accuracy and scalability.

5. Paper Title: Lights, Camera, Action! A Framework to Improve NLP Accuracy over OCR Documents

• What they did:
The study proposed a framework to improve the performance of NLP tasks (such as Named Entity Recognition) on OCR-generated text. It addresses the issue of OCR errors by generating synthetic noisy documents and training a model to restore corrupted text before applying NLP tasks.

• Technique used:
The system uses a document synthesis pipeline (Genalog) to generate degraded document images and corresponding OCR text. It introduces a text restoration model that predicts editing actions (INSERT, DELETE, REPLACE, INSERT_SPACE) instead of directly generating corrected text. The model uses character embeddings, convolutional layers, and sequence labeling techniques. OCR errors are measured using metrics like Character Error Rate (CER) and Word Error Rate (WER).

• **Limitation:**
The approach requires large amounts of synthetic training data and is computationally intensive. It may not perform well on extremely degraded documents and adds additional processing complexity to the pipeline. The system is also dependent on the quality of generated synthetic data and may not generalize perfectly to all real-world scenarios.

6. Paper Title: A Multi-View Framework to Detect Redundant Activity Labels for More Representative Event Logs in Process Mining

• What they did:
The study proposed a system to automatically detect redundant (duplicate) activity labels in event logs, where different labels represent the same real-world activity. This improves the quality of event logs and helps generate more accurate and understandable process models in domains like healthcare.

• Technique used:
The system uses a multi-view framework combining three perspectives: (1) control-flow similarity (based on ordering of activities), (2) attribute value similarity (categorical and numerical data comparison), and (3) semantic similarity (using NLP models). It applies Earth Mover's Distance (EMD) to compare distributions and uses a majority voting mechanism to combine results from all views and make the final decision.

• Limitation:
The approach is computationally complex due to multiple comparisons and models. It requires threshold tuning and may depend on data quality. Additionally, it is designed for structured event logs and may not directly apply to raw documents like PDFs without preprocessing steps.

7. Paper Title: LayoutLM: Pre-training of Text and Layout for Document Image Understanding

• What they did:
The study proposed a multimodal model (LayoutLM) that improves document understanding by combining textual content, layout (spatial position), and visual information from document images. It is designed to handle complex documents like forms, invoices, and receipts by understanding both the text and its placement on the page.

• Technique used:
The model is based on the BERT Transformer architecture and extends it by adding 2D layout embeddings (capturing word positions using bounding box coordinates) and image embeddings (using Faster R-CNN). It is pre-trained using tasks such as Masked Visual-Language Modeling (MVLM) and multi-label document classification, allowing it to learn relationships between text, layout, and visual features.

• Limitation:
The model requires large-scale datasets and high computational resources for training. It is complex to implement and not suitable for small-scale or beginner-level projects. Additionally, it depends on accurate OCR outputs, as errors in text extraction can impact performance.

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5. Proposed System

The system follows a hybrid approach combining OCR, NLP, machine learning, and rule-based validation.

System Pipeline:

1. **PDF Upload**
2. **PDF to Image Conversion**
3. **OCR (Text Extraction)**
4. **Text Preprocessing**
5. **Feature Extraction (TF-IDF)**
6. **Machine Learning Classification**
7. **Rule-Based Validation**
8. **Final Decision (Valid / Invalid)**

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6. Methodology

6.1 OCR (Text Extraction)

- Tools: EasyOCR / Tesseract
- Converts PDF/image into machine-readable text

6.2 Text Preprocessing

- Convert to lowercase
- Remove punctuation and stopwords
- Clean OCR noise

6.3 Feature Extraction

- Use TF-IDF (Term Frequency–Inverse Document Frequency)
- Converts text into numerical vectors

6.4 Machine Learning Model

- Algorithms: Logistic Regression / SVM
- Trained to classify:
  - Prescription
  - Non-prescription

6.5 Rule-Based Validation (Important Layer)

Checks presence of keywords:

Valid Indicators:

- Doctor name (Dr.)
- Hospital name
- Blood, units, transfusion

Invalid Indicators:

- UID, Government of India
- Address-heavy documents

6.6 Final Decision Logic

- If ML prediction = valid AND rules satisfied => ACCEPT
- Else => REJECT

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7. System Architecture

User Upload PDF

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PDF → Image Conversion
↓
OCR (Text Extraction)
↓
Text Cleaning
↓
TF-IDF Feature Extraction
↓
ML Classification
↓
Rule-Based Validation
↓
Final Output (Valid / Invalid)
↓
Trigger Donor Matching

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8. Technologies Used

- Python
- EasyOCR / Tesseract OCR
- Scikit-learn (ML models)
- pdf2image (PDF processing)
- NLP techniques (TF-IDF)

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9. Expected Outcomes

- Accurate validation of uploaded documents
- Prevention of misuse of the platform
- Automated and scalable verification system
- Improved trust and efficiency in blood donation process

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10. Future Scope

- Integration of deep learning models (e.g., LayoutLM)
- Layout-based document understanding

- Named Entity Recognition (NER) for extracting medical details
- Multi-language document support

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11. Conclusion

This project proposes a practical and scalable solution for validating medical prescriptions in a blood donation system. By combining OCR, NLP, machine learning, and rule-based validation, the system ensures that only genuine requests are processed, thereby improving reliability and preventing misuse.
